TARUN REDDY

Willing to relocate / open to visa sponsorship | +1 (351) 220-1157 | tarunreddy2811@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Machine Learning / AI Engineer with hands-on experience building end-to-end AI systems, from real-time computer vision pipelines to LLM-powered SQL agents and audio intelligence platforms. Co-founder of InsightScribe, a production-grade AI platform using OpenAI Whisper and LangChain. Strong foundation in model optimization, NLP, and deploying AI features via REST APIs.

EDUCATION

Master of Science in Computer Science — Clark University
Aug 2024 – May 2026
GPA: **3.7 / 4.0**

Bachelor of Technology in Computer Science — JNTU
Aug 2019 – May 2023
GPA: **3.5 / 4.0**

PROFESSIONAL EXPERIENCE

Teaching Assistant — Clark University

Aug 2025 – Present | Worcester, MA

- Mentored graduate students in machine learning concepts, Python environments, and full-stack integration
- Assisted with debugging ML pipelines, dependency issues, and model execution workflows
- Explained deep learning and system design concepts clearly to diverse technical audiences
- Designed and graded ML-focused assignments, providing detailed feedback on model implementation and evaluation approaches

AI/ML Engineering Intern — IMAI Innovations

Dec 2023 – May 2024 | Abu Dhabi, UAE (Remote)

- Optimized deep learning model architecture and training workflows, reducing training time by 20% while maintaining inference accuracy across evaluation metrics
- Built ML-driven HR automation tools that reduced manual PeopleOps workload by **40%**
- Engineered end-to-end ML pipelines including data preprocessing, feature engineering, model training, and evaluation
- Deployed AI-driven features via **secure REST APIs**, enabling seamless integration with client-facing platforms
- Evaluated model performance using precision, recall, F1-score, and ROC-AUC to ensure reliability
- Balanced model complexity, latency, and accuracy trade-offs to optimize deployment in resource-constrained environments.

PROJECTS

InsightScribe - AI Intelligence Platform | GitHub

(Co-Founder & ML/AI Engineer | In Development)

- Built audio transcription and NLP pipeline using **OpenAI Whisper and HuggingFace**, processing uploaded audio into structured, searchable transcripts
- Engineered cross-recording intelligence system to surface patterns and insights across multiple audio sessions
- Designed and integrated summarization and insight extraction features using **LLM-based pipelines**
- Collaborated in a co-founder engineering environment, owning the full ML/AI layer of a production-grade platform

Real-Time Facial Emotion & Age Detection System | Demo

(Independent ML Project | End-to-End Ownership)

- Built a real-time computer vision application using OpenCV, DeepFace, and Python, integrating live webcam feeds with deep learning models for facial emotion and age detection
- Engineered low-latency video stream processing to handle continuous frame input, achieving real-time inference with minimal frame drop
- Integrated a custom **CNN-based age prediction model** into the application layer, improving classification accuracy by **18%** through preprocessing optimizations
- Designed modular, maintainable code structure separating video capture, model inference, and UI overlay, demonstrating production-oriented software engineering practices.

EnterpriseIQ - LangChain SQL AI Agent | Demo

(Independent Project)

- Built a production-style **LLM-powered SQL agent** using **LangChain and Groq's LLaMA-3.3-70b**, enabling natural language querying of enterprise databases without SQL knowledge
- Engineered agent reasoning pipeline using LangChain's `create_sql_agent` with automatic **schema introspection**, handling complex multi-table JOINs and aggregations autonomously
- Designed prompt flow and tool-calling logic to resolve ambiguous business questions into accurate, context-aware SQL queries
- Built interactive **Streamli** interface with suggested query prompts, delivering a non-technical user-facing AI product end-to-end

SKILLS

Programming: Python, SQL, C, R
ML & AI: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, NLP, Computer Vision, CNNs, LSTMs, LLMs, LangChain
Data & Tools: Pandas, NumPy, Jupyter Notebook
Engineering: REST APIs, Git, Modular Code, Model Evaluation, Model optimization
Cloud: AWS, Google Cloud

CERTIFICATIONS

AWS Educate Machine Learning | Applied Data Science (IBM) | Data Analytics on Google Cloud